

Team 63: An Explainable Approach to Predicting NBA Game Winners

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ABSTRACT

This report presents an explainable machine learning model for predicting NBA game winners. Unlike betting site algorithms that are accurate but opaque, our approach emphasizes transparency using intuitive features such as home-court advantage, recent performance, and rest days. Predictions are benchmarked against betting odds and supported by visual explanations, creating an educational and trustworthy tool for understanding data-driven sports outcomes.

INTRODUCTION

Current sports prediction systems, such as betting site algorithms, achieve high accuracy but reveal little about how outcomes are determined. This lack of transparency limits fan engagement and trust. Our project introduces a transparent model that prioritizes interpretability over complexity. Using intuitive variables such as home-court advantage, rest days, and team performance, we show how each factor contributes to predictions. Evaluating against betting odds and explaining results visually will make NBA analytics more accessible and reliable for non-experts.

TARGET AUDIENCE, IMPACT & EVALUATION

The primary audience is basketball fans who want to know why a team is favored, not just who might win. Secondary audiences include sports analysts and students exploring explainable AI. If successful, the model will promote transparency and help fans interpret data-driven insights rather than rely on opaque odds. Success will be measured both quantitatively (accuracy vs. betting odds across seasons) and qualitatively (clarity and usefulness of explanations). Evaluation checkpoints at midterm and final stages ensure measurable progress toward an interpretable, trustworthy system.

RISK & PAYOFFS

Potential risks include lower predictive accuracy than professional systems and weak correlations among intuitive features. However, the benefits outweigh these limitations: even if slightly less accurate, the model will be transparent, educational, and free, supporting fairness and understanding in sports analytics.

The project requires no funding. All tools and data are open-source, including Python, SQLite, and public NBA datasets.

PLAN OF ACTIVITIES

The project will take six weeks, divided into three phases: Phase 1 (Weeks 1–2): Collect and clean data; validate features. Phase 2 (Weeks 3–4): Build interpretable models; evaluate vs. random baseline (>50% accuracy). Phase 3 (Weeks 5–6): Test across multiple seasons; compare to betting odds; refine visual explanations. For the midterm checkpoint we plan to produce a working single-season model with initial feature-impact visuals. Final Checkpoint: Multi-season model matching betting benchmarks and clearly explaining result.

ACTIVITY	PLAN WEEK	PLAN	ACTUAL WEEK	ACTUAL	PERCENT	WEEK									
	START	DURATION	START	DURATION	COMPLETE	7	8	9	10	11	12	13	14	15	
Research topic	7	1	7	1	100%										
Proposal Document	8	1	8	1	100%										
Role assignments	9	0.2	9	0.2	100%										
Data collection and cleaning	9	0.4	9	0.4	100%										
Data validation and normalization	9	0.4	9	0.4	100%										
Feature engineering	10	1	11	1	100%										
Model training and tuning	11	0.5	11		0%										
Initial evaluation	11	0.5	11		0%										
Testing and refinement	12	1	12		0%										
Data visualization	13	1	10	1	80%										
Final report and presentation	14	1	14		0%										

RESPONSIBILITIES

The following team members have contributed a similar amount of effort on all tasks:

- Oluwatosin Nicholas Adeyemi
- Omer Braude
- Julio Cesar Pujada Espinoza

EXPECTED INNOVATIONS

1. Momentum – Based Performance Features

We take on from resource materials, and put into action suggested new features: introduce momentum metrics that capture a team’s short-term performance trends, specifically the number of wins in the last 1, 3, 5, and 10 games. These rolling windows quantify recent form and consistency, allowing the model to detect shifts in team performance that static season averages miss.

2. Rest-Day Impact Modeling

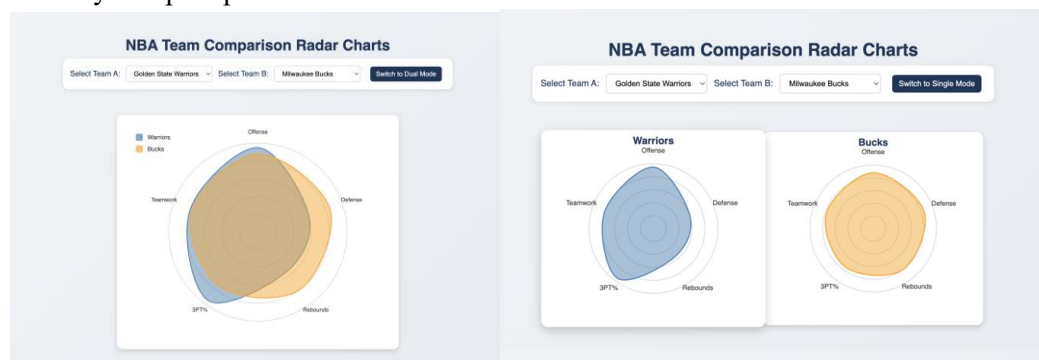
A rest feature measures the number of days since each team’s last game, directly integrating fatigue and recovery effects into the prediction pipeline. This reflects real-world scheduling dynamics, where back-to-back games or extended breaks can significantly affect outcomes.

3. Dual/Single Mode Radar Visualizations

We plan to implement interactive radar charts that visualize team strengths across selected features.

- **Dual Mode** will overlay both teams on one chart for direct comparison.
- **Single Mode** presents individual team profiles for side-by-side clarity and comparison.

The toggle button between both modes enhances interpretability and user engagement by allowing fans to easily compare performance traits.



Dual Mode vs Single Mode Comparisons

4. Transparent, Fan-Friendly Interpretability

Beyond prediction accuracy, our system emphasizes visual transparency. Each radar visualization directly reflects feature weights and model outputs, allowing users to visually understand why one team is favored, making explainable AI accessible to non-technical audiences.

UPCOMING EXPERIMENTS

1. Experiment Design

- a. **Analysis unit:** Each observation in our analysis corresponds to a single game played in the National Basketball Association (NBA).
- b. **Population:** All official NBA games since the inaugural 1946-47 season.
- c. **Filters:**
 - i. Include games from the “Regular season” and “Playoffs” only.
 - ii. Exclude Summer Season games, as these are experimental in nature and do not influence official season standings.
 - iii. Focus primarily on the last six seasons to capture current game dynamics, including changes in pace, scoring trends, and the increasing emphasis on mid-range and three-point shooting.
- d. **Variable use:** A time-lagged approach will be used to construct the dataset. To predict the outcome of a game at time t (Win = 1, Loss = 0 for the home team), we will utilize information from one or more past games ($t-1$ or earlier). This design ensures that only data available before the game is used for prediction, preserving temporal realism.

2. Assess prediction potential for new features

Evaluate individual prediction potential for each new feature, to assess their impact in the odds of winning for home team.

3. Address the evolving game dynamics by creating relative ratio features

To increase our final dataset volume, add more seasons into our analysis, and, to dilute the effect of the increasing total number of points scored -and other variables-, we will create Home vs Away ratios. For example, instead of “Home Total Points”, we will create “Percentage Ratio Home vs Away Total Points”. Intuitively, we would assume that if the Home team has scored 2x more points than the Away team in previous games, they will have a higher chance of winning.

4. Comparative Benchmarking

Our final experiment would be comparing our model’s accuracy to professional betting odds. Although our focus is on transparency rather than perfection, this benchmark will provide a clear indicator of the model’s real-world performance and relevance to existing prediction systems.

RELATED WORK

Existing research in sports prediction and explainable machine learning informs our project’s design. Studies on **model accuracy and calibration** show that accuracy alone is not sufficient to evaluate predictive quality. Walsh and Joshi [1] demonstrate that calibration—how well predicted probabilities match actual outcomes—can be more meaningful than raw accuracy. Similarly, Ren and Susnjak [11] benchmark explainable soccer models against bookmaker odds, emphasizing probabilistic consistency over binary accuracy. These insights support our plan to compare predicted probabilities with professional betting odds using calibration curves and Brier scores. However, prior work does not address interpretability or focus on basketball, which our project aims to extend.

Research on **game-context factors** highlights variables that strongly influence outcomes. Cuesta et al. [2] review home-court advantage, while Esteves et al. [3] show that back-to-back games and fatigue reduce win probability. Yan et al. [6] and Costa et al. [12] confirm similar effects in professional and collegiate basketball, validating our inclusion of home/away and rest-day features. Aoki et al. [13] further show that randomness limits predictive ceilings in sports, providing theoretical justification for emphasizing interpretability over perfect accuracy. Prior studies, however, focus narrowly on physical or situational effects; our work generalizes these variables statistically across multiple seasons to enhance clarity for fans. Early **NBA prediction models** [4], [5], [6] achieved modest accuracy improvements but paid little attention to explainability. More recent studies by Ouyang et al. [14] and He and Choi [15] employ ensemble and gradient-boosting models with SHAP analysis for NBA games, demonstrating that modern interpretable methods can rival complex predictors. Wang et al. [16] similarly apply explainable AI to analyze playing styles and match dynamics. While these works show strong performance, their explanations remain technical; we aim to simplify interpretability for general audiences through intuitive visualizations. Finally, **explainable AI in sports** demonstrates that interpretability and accuracy can coexist across domains. Lalwani et al. [7] applied explainable models with SHAP and ProtoDash in volleyball, while Chen et al. [8] used interpretable gradient boosting in soccer. Bunker and Thabtah [9] emphasize the need for transparency, Ribeiro et al. [10] introduce the model-agnostic LIME framework, Obradović and Kečo [17] review ML methods for sports prediction, and Odet et al. [18] combine prediction and explainability in rugby. These studies confirm that transparency enhances understanding, though most focus on single sports or datasets. Our project advances this by applying explainable modeling at a season-level NBA scale with simplified, fan-friendly insights.

REFERENCES

- [1] C. Walsh and A. Joshi, "Machine learning for sports betting: should model selection be based on accuracy or calibration?," arXiv preprint arXiv:2303.06021, 2023.
- [2] J. Cuesta et al., "Home advantage in professional basketball: crowd and travel effects," *J. Sports Analytics*, vol. 10, pp. 201–213, 2024.
- [3] J. Esteves et al., "The impact of rest days on basketball performance," *Eur. J. Sport Sci.*, vol. 21, pp. 599–610, 2021.
- [4] D. Miljković et al., "The use of data mining for basketball matches outcomes prediction," *Proc. IEEE Symp. Information Technology in Sports*, 2010.
- [5] H. Nguyen et al., "NBA game prediction using machine learning techniques," *Int. J. Data Sci.*, 2021.
- [6] R. Yan et al., "Determinants of home-team success in the NBA: a statistical analysis by quarters," *Sports Analytics Rev.*, 2024.
- [7] A. Lalwani et al., "Machine learning in sports: a case study on using explainable models for predicting outcomes of volleyball matches," arXiv preprint arXiv:2206.09258, 2022.
- [8] W. Chen et al., "Explainable gradient boosting for soccer outcome prediction," *Data Mining in Sports*, 2022.
- [9] R. Bunker and F. Thabtah, "A machine learning framework for sport result prediction," *Appl. Computing and Informatics*, vol. 15, no. 1, pp. 27–33, 2019.
- [10] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016.
- [11] R. Ren and A. Susnjak, "Predicting football match outcomes with explainable machine learning," arXiv preprint arXiv:2211.15734, 2022.
- [12] T. Costa et al., "A machine learning approach using interpretable models for NCAA basketball," *Proc. ICML Applications Conf.*, 2024.
- [13] R. Y. S. Aoki, R. M. Assunção, and P. O. S. V. de Melo, "Luck is hard to beat: the difficulty of sports prediction," arXiv preprint arXiv:1706.02447, 2017.
- [14] J. Ouyang et al., "Integration of machine learning XGBoost and SHAP for NBA game prediction," *Frontiers in Sports Analytics*, 2024.
- [15] L. He and J. Choi, "Stacked ensemble model for NBA game outcome prediction," *Sci. Rep.*, vol. 15, no. 1, pp. 1–12, 2025.
- [16] Z. Wang et al., "Explainable AI techniques with application to NBA analytics," *Neurocomputing*, vol. 520, pp. 32–41, 2022.
- [17] A. Obradović and N. Kečo, "Sports results prediction model using machine learning," *Sci. Appl. Res. J.*, vol. 73, pp. 184–189, 2024.
- [18] F. Odet et al., "A machine learning and explainability-driven methodology: a rugby case study," *Comput. Sports Sci.*, 2025.